

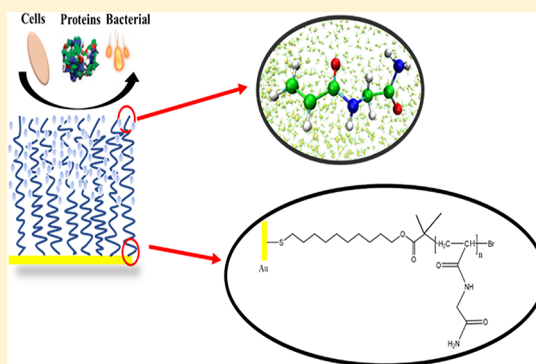
Synthesis and Characterization of Ultralow Fouling Poly(*N*-acryloyl-glycinamide) Brushes

Fengyu Yang,^{†,‡,§} Yonglan Liu,^{‡,§} Yanxian Zhang,[‡] Baiping Ren,[‡] Jianxiong Xu,^{*,†,‡} and Jie Zheng^{*,†,‡}

[†]Hunan Key Laboratory of Biomedical Nanomaterials and Devices, College of Life Science and Chemistry, Hunan University of Technology, Zhuzhou 412007, P. R. China

[‡]Department of Chemical & Biomolecular Engineering, The University of Akron, Akron, Ohio 44325, United States

ABSTRACT: The rational design of biomaterials with antifouling properties still remains a challenge, although this is important for many bench-to-bedside applications for biomedical implants, drug delivery carriers, and marine coatings. Herein, we synthesized and characterized poly(*N*-acryloylglycinamide) (polyNAGA) and then grafted poly(NAGA) onto Au substrate to form polymer brushes with well-controlled film stability, wettability, and thickness using surface-initiated atom transfer radical polymerization (SI-ATRP). The NAGA monomer integrates two hydrophilic amides on the side chain to enhance surface hydration, which is thought as a critical contributor to its antifouling property. The antifouling performances of poly(NAGA) brushes of different film thicknesses were then rigorously assessed and compared using protein adsorption assay from undiluted blood serum and plasma, cell-adhesive assay, and bacterial assay. The resulting poly(NAGA) brushes with a film thickness of 25–35 nm exhibited excellent in vitro antifouling ability to prevent unwanted protein adsorption ($<0.3 \text{ ng/cm}^2$) and bacterial and cell attachments up to 3 days. Molecular dynamics (MD) simulations further showed that two hydrophilic amide groups can interact with water molecules strongly to form a strong hydration layer via coordinated hydrogen bonds. This confirms a positive correlation between antifouling property and surface hydration. In line with a series of polyacrylamides and polyacrylates as antifouling materials synthesized in our lab, we propose that small structural changes in the pendent groups of polymers could largely improve the antifouling capacity, which may be used as a general design rule for developing next-generation antifouling materials.



INTRODUCTION

Biofouling prevention from unwanted biomolecule and micro-organism attachment on the artificial surfaces places high demands on many biomedical and industrial applications,^{1–9} such as biomedical implants,^{10,11} drug delivery carriers,¹² biosensors,^{13,14} and marine coating.¹⁵ For biomedical usage,^{16–19} even $\sim 5 \text{ ng/cm}^2$ of fibrinogen adsorption on the surgical implants will trigger very complex foreign body responses, including a series of blood platelet adhesion, complement and leukocyte activation, tissue fibrosis, thrombosis coagulation, and bacterial infection.^{20–22} For industrial usage,²³ microorganism attachment and proliferation on marine hulls annually cost millions of dollars for ship maintenance and environmental protection.

Immobilization of hydrophilic or zwitterionic polymers^{24–26} on different surfaces is a common strategy to create antifouling surfaces to prevent biofouling. It is generally accepted that polymer coatings offer a strong surface hydration barrier and steric repulsion to prevent the unwanted adsorption of foulants.^{27–30} Hydrophilic polymer coatings (PEG,^{31,32} polysaccharides,^{33,34} polyacrylate,^{21,35} and polyamide^{36,37}) achieve surface hydration via hydrogen bonding of polymers with water molecules, whereas zwitterionic polymer coatings (poly(carboxybetaine methacrylate),³⁸ poly(sulfobetaine methacry-

late),³⁹ and poly(3-(1-(4-vinylbenzyl)-1*H*-imidazol-3-ium-3-yl)propane-1-sulfonate)²⁸) achieve electrostatically induced surface hydration where water molecules bind to zwitterionic groups even stronger. The formation of a strong hydration layer at the polymer/water interface offers an energetic barrier to effectively inhibit unwanted foulant adsorption on polymer coatings. The strength of surface hydration is mainly determined by the intrinsic physicochemical properties of polymers (i.e., molecular structure, charge distribution, and hydrophobicity) and the extrinsic coating properties (i.e., packing density, film thickness and roughness, and chain conformation).⁴⁰

We have previously demonstrated that a family of polyacrylamide and polyacrylate brushes, including poly(*N*-hydroxyethyl acrylamide) (poly(HEAA)),³⁷ poly(*N*-acryloyl-aminoethoxyethanol) (poly(AAEE)),⁴⁰ poly(2-hydroxyethylmethacrylate) (poly(HEMA)),⁴¹ poly(*N*-hydroxymethylacrylamide) (poly(HMAA)),⁴² poly(*N*-hydroxypropylacrylamide) (poly(HPAA)),⁴² and poly(*N*-hydroxypentyl acrylamide) (poly(HPenAA)),⁴² exhibited high surface resistance to the

Received: October 1, 2017

Revised: November 18, 2017

Published: November 21, 2017

unwanted adsorption of proteins, cells, and bacteria under optimal coating conditions. We observe that small structural changes in the polymers could largely enhance antifouling capacity and surface hydration, both of which show a positive correlation. Specifically, inclusion of amide ($-\text{CONH}-$) or ethylene glycol ($-\text{CH}_2\text{CH}_2-\text{O}-$) pendant groups and the shorter carbon space length ($\text{CSL} = 1$ or 2) between two hydrophilic groups enables great enhancement of surface hydration and antifouling property. On the basis of this structural-design principle, herein we synthesized a hydrophilic *N*-acryloylglycinamide (NAGA) monomer containing two amide groups with one CSL between them⁴³ and then grafted and polymerized NAGA on the Au substrate to form poly(NAGA) brushes with different film thicknesses using the surface-initiated atomic transfer radical polymerization (SI-ATRP).⁴⁴ The two amide groups in poly(NAGA) are expected to strongly coordinate with water molecules to form a tight water layer at the polymer brush interface, which in turn induces a better antifouling property. Consequently, we examined the antifouling property of poly(NAGA) brushes using different protein, cell, and bacterial assays. We also examined the effect of film thickness as a key factor for antifouling performance. Collective experimental results showed that at a film thickness of 25–35 nm, poly(NAGA) brushes highly resisted nonspecific protein adsorption of <0.3 ng/cm² (0.3 and 5 ng/cm² of nonspecific protein adsorption are considered as threshold values of nonfouling and low fouling, respectively^{40,45,46}) from undiluted blood plasma and serum and cell and bacterial attachments. Molecular dynamics (MD) simulations were also performed to study the structure and dynamics of water molecules near the polymer interface and to correlate with the experimental observation. By identifying poly(NAGA) as antifouling materials, the pool of antifouling polymers is further expanded.

MATERIALS AND METHODS

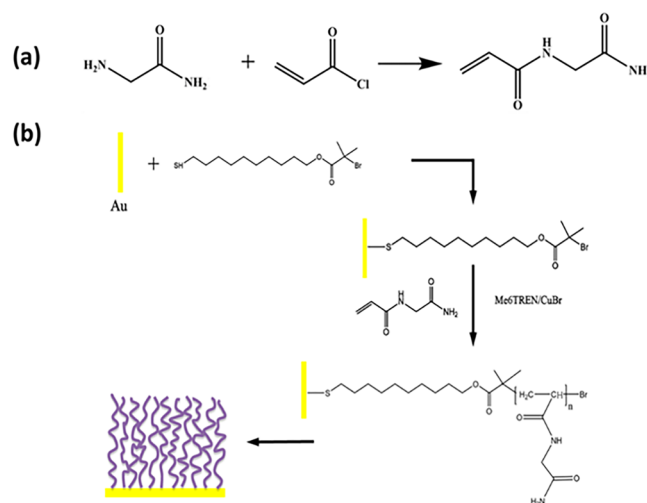
Materials. 2-Propenoyl chloride (97.0%), copper(I) bromide (CuBr, 99.999%), potassium carbonate, ethanol (absolute 200 proof), 2,2,2-trifluoro-1-ethanol, acetone, diethyl ether (absolute), and phosphate buffered saline (0.01 M PBS, pH = 7.4, 0.0027 M KCl and 0.138 M NaCl) were purchased from Sigma-Aldrich Inc. (Milwaukee WI). Glycinamide hydrochloride was purchased from Matrix Scientific (Columbia, SC). Tris[2-(dimethylamino)ether]amine (Me₆TREN) was obtained from Alfa Aesar Inc. (WI). Glycinamide hydrochloride was purchased from Matrix Scientific (Columbia, SC). Human blood serum and plasma were purchased from BioChemed Service (Winchester, VA). The water was processed using the Millipore system and the resistivity is 18.2 MΩ·cm. The ATRP initiator, ω -mercaptoundecyl bromoisobutyrate and bovine aortic endothelial were kindly provided by Prof. Shaoyi Jiang at the University of Washington. The ATRP initiator was synthesized by the reaction of bromoisobutyryl bromide and 11-mercapto-1-undecanol as reported in previous work.⁴⁷

Synthesis of *N*-Acryloylglycinamide (NAGA). NAGA monomer was synthesized by the reaction of glycinamide hydrochloride and 2-propenoyl chloride.⁴³ Glycinamide hydrochloride (6.30 g, 0.05706 mol), cold deionized water (6 mL), 2 mol L⁻¹ cool potassium carbonate (33.6 mL), and cold diethyl ether (18 mL) were mixed together into a 100 mL three-necked glass flask at 0 °C in an ice bath with a thermometer. Next, a solution of 2-propenoyl chloride (5.90 g, 0.065 mol) and 24 mL diethyl ether was added to the mixture solution drop-wise. The mixture solution was magnetically stirred at 0 °C for almost 1 h and then was further stirred for 4 h at room temperature to complete the reaction. Subsequently, 6 mol L⁻¹ HCl was used to achieve a pH of 2. To remove the organic phase, the mixture solution was washed with 150 mL of diethyl ether three times. Next, the rotary

evaporation was used to remove the remaining diethyl ether in the mixture. Again, 2 mol L⁻¹ NaOH was used to adjust the pH to neutral and the mixture solution was freeze-dried to remove the water phase. The well freeze-dried raw product was washed three times with 150 mL of ethanol/methanol mixture (4/1, v/v). Then, the mixed solvents of ethanol and methanol were removed by rotary evaporation. The concentrated mixture was recrystallized at 0 °C to obtain the solid NAGA monomer. Then, the NAGA monomer was filtrated and dried in vacuum to remove the diethyl ether attached on the NAGA monomer surface. The yield was ~65%. ¹H NMR (300 MHz, D₂O) (ppm) was used to check and confirm the properties and purity of the product.

SI-ATRP of NAGA on Surface Plasmon Resonance (SPR) Chips. SPR chips include a 2 nm adhesion-promoting chromium layer and a 48 nm gold layer on a microscope glass slide.⁴¹ To avoid side effects, the golden coated SPR chip was washed with hydrophilic solvent acetone, ethanol, and water three times separately to remove the impurities attached on the bare golden surface. The chips were then sterilized under a UV zone for 20 min, washed with pure acetone, ethanol, and DI water three times, and were totally air-dried before use. The precleaned Au substrates were soaked in ω -mercaptoundecyl bromoisobutyrate (initiator)/ethanol (1 mM:1) overnight to obtain the well-prepared initiator self-assembled monolayer (SAM). Then, the chips were washed with ethanol and pure water and dried in a nitrogen stream before use. Scheme 1 describes a general strategy for

Scheme 1. (a) Synthesis of NAGA Monomer and (b) Schematic SI-ATRP Process for Grafting Poly(NAGA) onto an Initiator-Coated Gold Surface



grafting poly(NAGA) onto gold surfaces covered with initiators via SI-ATRP. CuBr (16.8 mg, 0.170 mmol) and SPR gold chips coated with SAMs were placed in the first reaction tube sealed with rubber spectrum stoppers for 20 min under nitrogen injection and vacuum degassing for three cycles. In the second tube, NAGA (0.3 g, 3.21 mmol), Me₆TREN (40 mg, 0.174 mmol), Milli-Q water (2.5 mL), and 2,2,2-trifluoro-1-ethanol (3 mL) were mixed together. The mixture was well-mixed by magnetic stirring under nitrogen protection to remove oxygen from the reaction system. Then, the mixed solution was transferred using a syringe to the tube with a chip and CuBr under nitrogen protection. Subsequently, the first tube was shaken for 20 min to mix well. Different film thicknesses can be controlled by polymerization times, 2,2,2-trifluoro-1-ethanol/water ratios, and monomer concentrations.⁴¹ After grafting poly(NAGA) on the SPR substrates, the chips were soaked in PBS for 3 days to remove possible unbounded and loosely bounded polymers.

Film Thickness of Poly(NAGA) Brushes by Ellipsometry. The alpha-SE ellipsometry (J.A. Woollam Co., Lincoln, NE) was used to measure the film thicknesses of poly(NAGA) brushes on the Au surface with a He–Ne laser ($\lambda = 632.8$ nm) at a fixed angle of

incidence of 70°. A refractive index of 1.45 was assigned to the polymer layers. Three different locations of each sample were chosen to obtain the mean film thickness and associated standard deviation. Herein, different film thicknesses of poly(NAGA) can be tuned by controlling different ATRP reaction conditions.⁴⁰

Protein Adsorption of Poly(NAGA) Brushes by SPR. Protein adsorption was measured by a custom-built four-channel SPR sensor based on wavelength interrogation.¹⁶ Four independent parallel channels were used to deliver the liquid sample during the experiment. A peristaltic pump (Ismatec) was also utilized to deliver degassed PBS or undiluted plasma and serum in four channels. After being rinsed with PBS for 3 days, the SPR chips immobilized with poly(NAGA) brushes were dried in air and then attached to the base of the prism; a baseline signal was obtained by flowing degassed PBS buffer (1 mg/mL) for 10 min at a rate of 0.05 mL/min through the sensor until the baseline is stable.³⁷ For each poly(NAGA) brush coated on the SPR chip, undiluted (100%) blood serum and undiluted blood plasma were made to flow through independent channels for 10 min to obtain an initial protein adsorption amount. Then, PBS solution was used to remove the unbound or loosely bound proteins. Protein adsorption was quantified by measuring wavelength change between the baselines before and after protein adsorption. An SPR wavelength change of 1 nm can be converted to 15 ng/cm² of adsorbed protein at 750 nm. For a four-channel SPR sensor, one channel was used for the control experiment and the other three channels were utilized to measure protein adsorption on the same brush.

Contact Angle Measurements. The Ramé-hart goniometer (model 100-00, Mountain Lake, NJ) was used to test the water contact angles of different thicknesses of poly(NAGA) brushes. Three positions of each poly(NAGA) brush were chosen and reported as mean $\pm$ standard deviation.

Cell Adhesion Assay. The cell suspension was prepared by the following procedure.¹⁶ Bovine aortic endothelial cells (BAEC) were incubated in a tissue culture polystyrene (TCPS) flask in a humidified incubator with 5% CO₂ at 37 °C. The culture medium that provides supplementation for cell growth includes Dulbecco's modified Eagle's media (DMEM) containing fetal bovine serum (FBS) (10 wt %), sodium pyruvate (1 wt %), nonessential amino acids (1 wt %), and penicillin streptomycin solution (2 wt %). After about 7 days to reach full confluence, BAEC cells were removed from the flask using a normal passaging protocol and diluted by the medium to a concentration of 10⁵ cells/mL solution. Poly(NAGA) brushes were immersed in disinfected PBS (phosphate-buffered saline) for 72 h before use. Samples were then placed in a 6-well plate with 3 mL bovine aortic endothelial cell suspension. The cells were incubated for 72 h at 37 °C. The EVOS XL core inverted microscope (Advanced Microscopy Group, Bothell, WA) was used to observe the morphology and proliferation of cells on different surfaces with 10 times their original size. Then, different surfaces were washed with PBS. Five typical images for each sample were chosen, and the attached cell numbers were counted manually and reported as mean $\pm$ standard deviation.

Bacterial Assay. Two kinds of bacteria (*Escherichia coli* (Gram-negative) and *Staphylococcus epidermidis* (Gram-positive)) were chosen for the bacterial attachment test. The lysogeny broth (LB; BD, Franklin Lakes, NJ) was used to provide supplementations for *E. coli* (Gram-negative) and *S. epidermidis* (Gram-positive) in agar plates for 12 h at 37 °C. Several colonies of the each bacteria were chosen and incubated in LB medium (20 mL), separately. The bacterial solutions were shaken at 200 rpm for 8 h at 37 °C and were then diluted with the LB medium to achieve a final density of 10⁵ cells/cm². The poly(NAGA) brushes were sterilized by ethanol for 15 min. After that, poly(NAGA) brushes were rinsed with PBS for 15 min. Each 4 mL of a bacterial suspension was placed on a sterile plate with a chip for 12 h at room temperature. After washing with Mili-Q water twice, all brushes were stained with a Live/Dead BacLight kit (Thermo Fisher Scientific Inc., NY) before imaging. The morphology of each brush was visualized by an Olympus IX81 fluorescence microscope with a 40 $\times$ lens. The numbers of attached bacteria were finally counted from five typical images.

Molecular Dynamics Simulation of Poly(NAGA). The topological and the parameter information of the NAGA monomer were derived from the CGenFF ParamChem tool (<https://www.paramchem.org/>),^{48–51} whose force field parameters are compatible with the CHARMM format.⁵² The force field parameters of the NAGA monomer were further validated by a short 2 ns explicit-water MD simulation at 298 K. The intramolecular connectivity and topology were well maintained during MD simulations as compared to those obtained by Gaussian95. MD simulations of a 16-mer of poly(NAGA) in a TIP3P water box of 60 $\times$ 60 $\times$ 60 Å³ were conducted using the NAMD program⁵³ and CHARMM force fields.⁵⁴ Every simulation system was energy minimized by the 5000-step conjugate minimization method before MD simulations. A switch function with a twin cutoff of 12 and 14 Å was applied for the short-range van der Waals (VMD) interactions, whereas a force-shifted method with a 12 Å cutoff was applied to the long-range electrostatics interactions. The Langevin method was used to maintain a constant temperature of 298 K and a pressure of 1 atm for the NPT ensemble with three-dimensional periodic boundary conditions. A total of 40 ns MD simulations was run to examine the structure and dynamics of water molecules near the poly(NAGA).

RESULTS AND DISCUSSION

Surface Characterization of Poly(NAGA) Brushes. The NAGA monomer was synthesized using the reaction of glycinamide hydrochloride and 2-propenoyl chloride in diethyl ether (Scheme 1a). NAGA identity and purity were validated by ¹H NMR. ¹H NMR (D₂O): δ = 3.7 (H4, –NH–CH₂–), 5.6 (H2, CH₂=CH–), 6.1 and 6.3 (H1, 3, CH₂=CH–) ppm (Figure 1). The poly(NAGA) brush was prepared on a gold

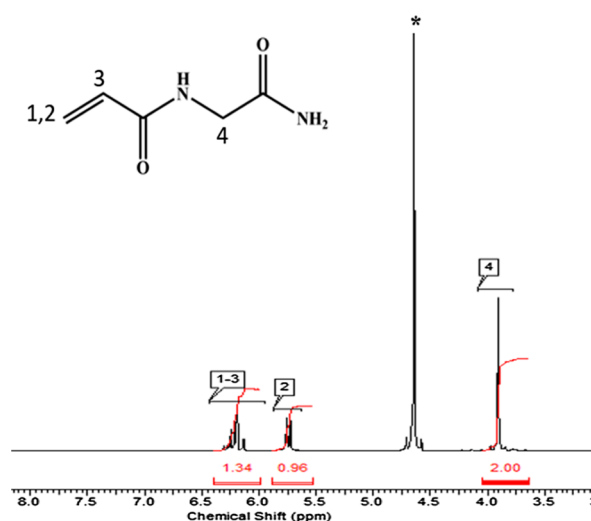


Figure 1. ¹H NMR spectra of NAGA monomer (300 MHz, * = solvent signal).

substrate via a two-step procedure (Scheme 1b), that is, first grafting the ω -mercaptoundecyl bromoisobutyrate SAM on the gold substrate, followed by the SI-ATRP to grow poly(NAGA) chains to form a brush.

Surface hydrophilicity, as measured by the water contact angle, is critical for the antifouling property. Figure 2a shows the water contact angle of a bare gold chip, an initiator-coated chip, and a poly(NAGA) brush-coated chip with a polymer thickness of $\sim$ 25 nm, which provides the step-by-step characterization for the surface hydration of the polymer-grafting surfaces by the SI-ATRP. It can be seen that the water contact angle changed from 54° on the bare-gold chip to 74° on the initiator-coated chip (mainly because of the presence of

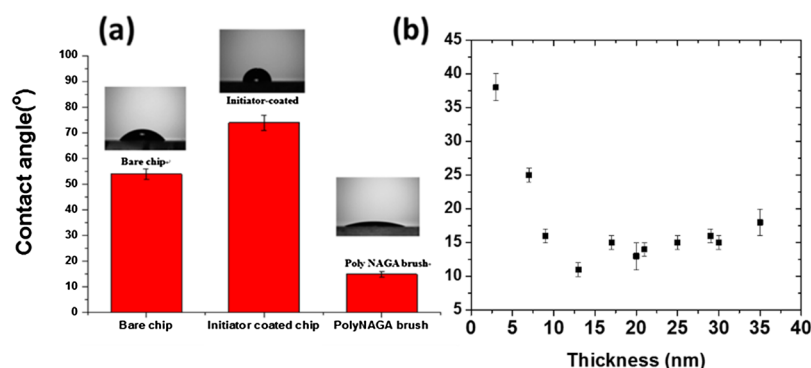


Figure 2. (a) Water contact angles on a bare chip, an initiator-coated chip, and a poly(NAGA) brush (~25 nm)-coated chip. Error bars represent the standard deviation of the mean ($n = 3$). (b) Water contact angle as a function of the film thickness of poly(NAGA) brushes.

hydrophobic Br atoms and alkyl groups) and to 15° on the poly(NAGA)-grafted chip. Clearly, upon grafting poly(NAGA) onto the initiator-immobilized golden surface, water contact angle decreased significantly, demonstrating that the poly(NAGA)-grafted surface displays hydrophilic character similar to other antifouling surfaces.

We further measured the dynamic changes of contact angles on poly(NAGA) brushes of varied film thicknesses from 3 to 36 nm. As shown in Figure 2b, except for a very thin brush with 3 nm film thickness, all poly(NAGA) brushes with different film thicknesses (7–35 nm) displayed hydrophilic character, as evidenced by small contact angles of $<26^\circ$. When film thickness slightly increased from ~ 3 to ~ 13 nm, water contact angles significantly decreased from $\sim 38^\circ$ to $\sim 11^\circ$. When film thickness further increased from ~ 13 to ~ 36 nm, water contact angles slightly increased from $\sim 11^\circ$ to $\sim 15^\circ$, indicating that surface hydrophilicity was not so sensitive to film thickness. Any poly(NAGA) brush with ~ 13 nm or higher film thickness enables the production of sufficient hydrophilic property. Visual inspection also showed that when water was dropped and spread on poly(NAGA) brushes, it needed approximately 5 s for both water and polymers to reach an equilibrium state. As we can see in Figure 2, the contact angles increased with film thickness (thickness > 20 nm). This phenomenon could be explained by the “bridge effect”, in which the grafted polymer chains at the solvent–vapor interface generate partial wetting and nonzero surface pressure.^{55,56} Thus, with the increase of the polymer chain length, polymer chains become more flexible and gain more freedom, inducing the bridge effect and the increase of the contact angles. We should also note that although polymer-coated antifouling surfaces are hydrophilic, most of these hydrophilic surfaces are not sufficient enough to inhibit the unwanted adhesion of cells, bacteria, or other microorganisms at the ultralow fouling level.

Protein Adsorption on Poly(NAGA) Brushes. To quantify the antifouling property of the surface in terms of the amount of nonspecific protein adsorption, it is generally accepted that the surface that adsorbs 5 ng/cm^2 proteins is defined as the superlow fouling surface, whereas the surface that adsorbs 0.3 ng/cm^2 proteins is defined as the nonfouling surface. Our previous and other studies have shown that the use of single-protein solution is not a reliable way to truly determine the antifouling property of the surfaces. Thus, we used both undiluted human blood plasma and serum that contain ~ 500 different proteins to test the poly(NAGA) brushes of different film thicknesses. As a proof-of-concept, Figure 3 shows two typical SPR sensorgrams to determine

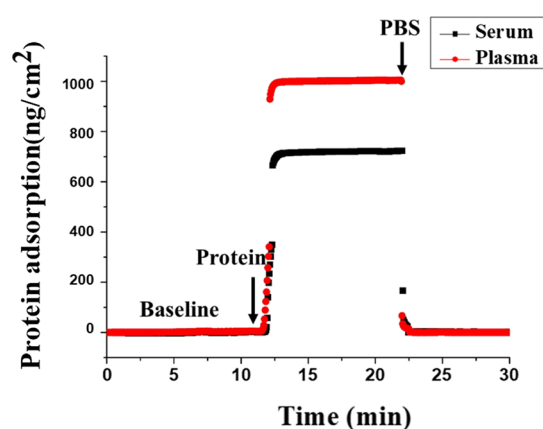


Figure 3. SPR sensorgrams to show protein adsorption on a poly(NAGA) brush of ~ 25 nm thickness from undiluted human blood plasma and serum.

adsorbed proteins on the poly(NAGA) brush of 25 nm. In Figure 3, upon flowing undiluted human blood plasma and serum through different SPR channels coated with a poly(NAGA) brush, the initial adsorption amounts of proteins were extremely high ~ 988 and 687 ng/cm^2 from undiluted human blood plasma and serum, respectively. However, when switching the plasma/serum solutions to the PBS solution, the adsorbed proteins on poly(NAGA) brushes were almost entirely removed from the surface by PBS, leading to the surface being free from proteins ($<0.3 \text{ ng/cm}^2$ adsorbed proteins), indicating that initial adsorbed proteins were weakly attached to poly(NAGA) brushes.

Furthermore, we examined the effect of the film thickness of poly(NAGA) brushes on protein adsorption from undiluted human blood plasma and serum. In the case of human blood plasma tests, Figure 4a shows that at a thin film thickness ranging 3–18 nm, poly(NAGA) brushes adsorbed a large amount of plasma proteins of $8\text{--}43 \text{ ng/cm}^2$. However, at a film thickness ranging 25–35 nm, poly(NAGA) brushes almost did not adsorb any protein, achieving a nonfouling level of $<0.3 \text{ ng/cm}^2$. In Figure 4b, poly(NAGA) brushes showed a similar protein adsorption trend from undiluted blood serum, that is, poly(NAGA) brushes with too small or too large film thicknesses induced large protein adsorption. For the thick polymer brush, the polymer chains entangle together and this causes the increase of the interchain interaction and the decrease of hydrogen bond interactions with water, leading to weak surface hydration to resist protein adsorption. For the

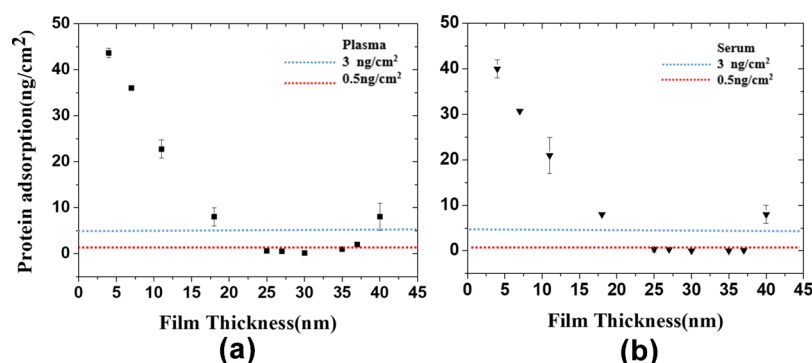


Figure 4. Effect of film thickness (3–42 nm) of poly(NAGA) brushes on protein adsorption from undiluted human blood (a) plasma and (b) serum.

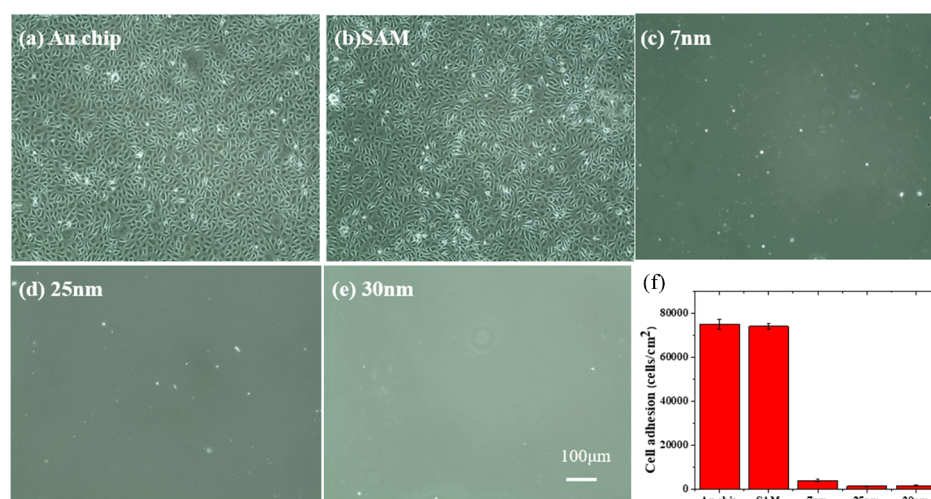


Figure 5. Representative microscopy images of BAEC cell attachment on (a) pure Au chip, (b) initiator-coated chip, and poly(NAGA) brushes with film thickness of (c) ~ 7 , (d) ~ 25 , and (e) ~ 30 nm after 72 h culture with cells. (f) Statistical analysis of the adhered BAEC cell density on the surfaces above ($n = 5$).

thin polymer brush, short chains cannot form strong surface hydration with water, leading to large protein adsorption.⁴¹ Only at an optimal film thickness ranging 25–35 nm, poly(NAGA) brushes showed undetectable nonspecific protein adsorption (<0.3 ng/cm²). As shown in Figure 4, large error bars only appeared in polymer brushes with very thin (<15 nm) or very thick (>40 nm) film thickness, simply because these polymer brushes may have varied surface roughness, which would in turn affect protein adsorption, leading to the relatively large error bar in few cases. On the other hand, this also explains that all superlow fouling brushes at a film thickness ranging 20–40 nm have very small errors (too small to be captured by the eyes) for protein adsorption owing to the smooth and uniform surface. Consistent with our previous and other studies, although protein adsorption from single-protein solution is generally insensitive to film thickness, there always exists an optimal film thickness to achieve minimum protein adsorption from complex biological media. It has been reported that the optimal film thickness of polymer brushes to achieve a superlow fouling level (5 ng/cm²) is 10–40 nm for poly(HEAA) brush,³⁷ 10–40 nm for poly(AAEE) brush,⁴⁰ 20–30 nm for poly(HEMA) brush,⁴¹ 27–33 nm for poly(VBIPS) brush,²⁸ 10–15 nm for poly(CBMA) brush,⁵⁷ and 5–12 nm for poly(SBMA) brush.²¹ Collective data from prior studies appear to suggest that 20–35 film thickness is the optimal range to reduce protein adsorption, because in this range, intra and intermolecular associations between polymer

chains are optimized to achieve high surface hydration for effectively preventing protein adsorption from the surface.

Cell Adhesion on Poly(NAGA) Brushes. Although the inhibition of initial nonspecific protein adsorption on the surface can greatly reduce the risk of subsequent biofilm formation, preventing protein adsorption from the surface does not necessarily, in most cases, prevent other biomolecule adsorption from the surface. A practical antifouling surface should be able to resist different unwanted biofoulant adsorption from real-world biological media containing cells and bacteria. Therefore, we further studied the adhesion of bovine aortic endothelial cells (BAECs) onto the poly(NAGA) brushes of different film thicknesses upon three day cell culture at 37 °C. As controls, optical images (10 $\times$ magnification) in Figure 5a,b showed that both pure Au surface and initiator-coated SAM surface were fully covered with BAECs, whose adhered cell densities reached full confluence of $7.4\text{--}7.5 \times 10^4$ cells/cm². For the poly(NAGA) brushes with a small thickness of 7 nm, a large number of cells were observed to adhere onto the thin poly(NAGA) brush (Figure 5c), resulting in an adhered cell density of $\sim 6.25 \times 10^3$ cells/cm² (Figure 5f). At the two optimal film thicknesses of 25 and 30 nm, almost no cells were attached on poly(NAGA) brushes (Figure 5d,e), confirming that poly(NAGA) brushes exhibit high surface resistance to both protein and cell adsorption at similar film thicknesses.

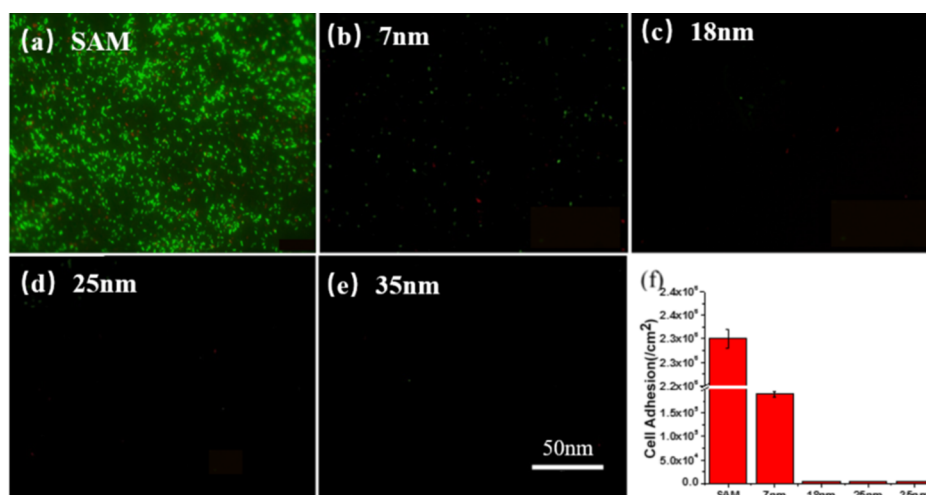


Figure 6. Representative fluorescence microscopy of *E. coli* attachment on (a) initiator-coated chip and poly(NAGA) brushes with film thickness of (b) ~7, (c) ~18, (d) ~25, and (e) ~35 nm after 12 h incubation. (f) Statistical analysis of adhered *E. coli* density on different surfaces ($n = 5$).

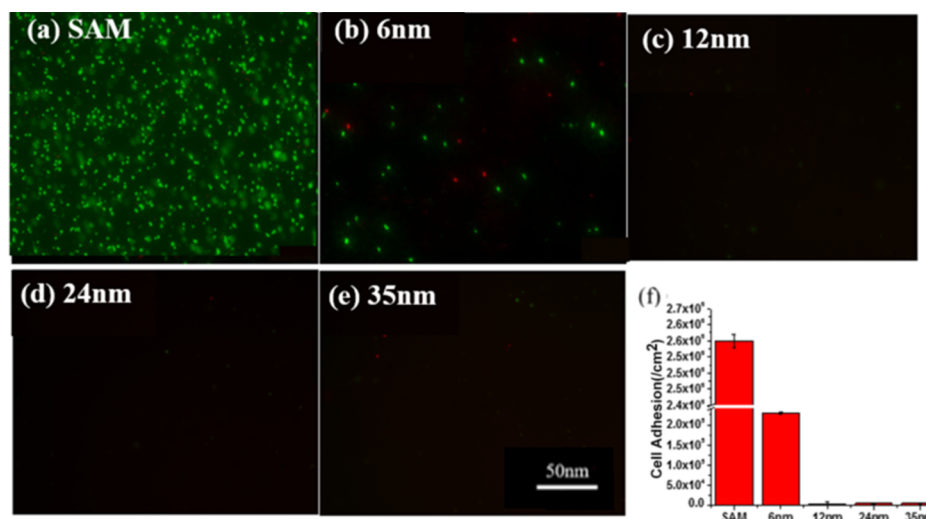


Figure 7. Representative fluorescence microscopy of *S. epidermidis* attachment on (a) initiator-coated chip and poly(NAGA) brushes with film thicknesses of (b) ~6, (c) ~12, (d) ~24, and (e) ~35 nm thickness after 12 h incubation. (f) Statistical analysis of adhered *S. epidermidis* density on different surfaces ($n = 5$).

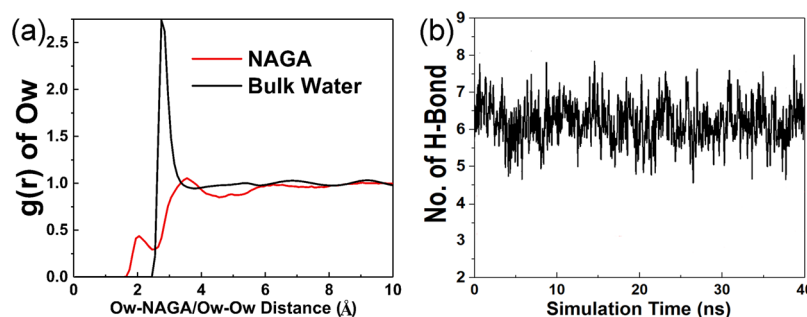


Figure 8. (a) Radial distribution function (RDF) profiles of water molecules around poly(NAGA) (red line) and bulk water (black line). (b) Average number of hydrogen bonds per NAGA monomer with water molecules.

Bacterial Adhesion on Poly(NAGA) Brushes. Finally, we challenged poly(NAGA) brushes with Gram-negative *E. coli* and Gram-positive *S. epidermidis* because natural selection enables bacteria to survive in a very harsh environment. Both *E. coli* and *S. epidermidis* are major virulence factors that induce fast biofilm formation,⁵⁸ and they are also known to have strong

drug resistance for many antibiotics (penicillin, amoxicillin, and methicillin).⁵⁹ Visual inspection in Figure 6 shows that upon 12 h incubation, a large amount of *E. coli* was attached to the initiator coated chip, whereas few *E. coli*, if any, were attached to all tested poly(NAGA) brushes. Quantitatively, the average *E. coli* densities on poly(NAGA) brushes of 7 and 18–35 nm

were 1.9×10^5 and 4.5×10^3 to 5.0×10^3 cells/cm², respectively, much less than $\sim 4 \times 10^6$ cells/cm² on initiator-coated chips (Figure 6f).

In Figure 7, a similar trend of *S. epidermidis* adsorption on different surfaces was observed. The surface density of adherent *S. epidermidis* on the poly(NAGA) brushes of 12, 24, and 35 nm was very small (4.5×10^3 to 5.0×10^3 cells/cm²) and did not change significantly with film thickness. Differently, a thin poly(NAGA) brush of 6 nm led to a relatively large adherence cell density of 2.3×10^5 cells/cm².

MD Simulations of Poly(NAGA) Hydration. MD simulation is a powerful tool for many different systems involving proteins,⁶⁰ polymers,⁴⁰ and inorganic or organic molecules.⁶¹ Hence, we performed all-atomic MD simulation to explore the dynamics properties of hydrated water associated with poly(NAGA) at the atomic level. Figure 8a displays the radial distribution functions (RDFs or $g(r)$) profiles for both water molecules around poly(NAGA) and bulk water. It can be seen that different from bulk water (the TIP3P model) that only displayed single large peak at 2.8 Å,⁶² poly(NAGA) displayed the first small hydration layer at 2.1 Å and the second large hydration layer at 3.6 Å, indicating that poly(NAGA) tends to form hydrogen bonds with water molecules in these two regions. After 4 Å, water molecules around poly(NAGA) showed density distribution similar to bulk water. Furthermore, we calculated the residence time of water molecules around poly(NAGA) within the first and second hydrogel shells of 2.1 and 3.6 Å. The residence time of water molecules within both hydrogel shells remained almost the same (29.4 ps), which was almost two-times longer than poly(HEAA) (13.5 ps) and poly(AAEE) (12.7 ps) in our previous study.⁴⁰ This further suggests that water molecules tend to stay longer with poly(NAGA) than with poly(HEAA) and poly(AAEE). We calculated the average number of hydrogen bonds between poly(NAGA) and water molecules, as normalized by the number of NAGA monomers. Figure 8b shows that each NAGA monomer consistently formed ~ 6.5 hydrogen bonds with water molecules during the entire 40 ns MD simulations, consistent with the long residence time of water molecules bound to NAGA molecules. Our MD simulations of residence time and hydrogen bonds of water molecules near poly(NAGA) chains demonstrate the strong binding between them, which are positively correlated with antifouling performance from experiments. These strongly bound water molecules form a strong energetic barrier to prevent protein adsorption and desolvation of the hydration layer around polymers is energetically unfavorable.

CONCLUSIONS

In this work, we synthesized and characterized a highly hydrophilic monomer of NAGA containing two hydrophilic amide groups and then polymerized and grafted NAGA monomers onto the Au substrate to form polymer brushes using the SI-ATRP. Collective data from SPR and water contact angle experiments revealed a strong dependence of the film thickness of poly(NAGA) brushes on their surface hydration and antifouling property. It was found that at the optimal film thickness of 25–35 nm, poly(NAGA) brushes exhibited not only high surface hydrophilicity as evidenced by the lowest water contact angle of $\sim 15^\circ$ but also high surface resistance to protein adsorption from undiluted protein plasma and serum at the ultralow fouling level (<0.3 ng/cm²). High surface hydrophilicity seems to positively correlate with high surface

resistance ability. It was further shown that poly(NAGA) brushes can effectively resist cell adhesion up to 3 days and bacterial attachment in 12 h. MD simulations confirm the formation of a hydration layer of poly(NAGA) that explains its antifouling property. We expect that simple structural modification of polyacrylamides and polyacrylates should be able to improve antifouling performance of the materials.

AUTHOR INFORMATION

Corresponding Authors

*E-mail: xujianxiong8411@163.com (J.X.).

*E-mail: zhengji@uakron.edu (J.Z.).

ORCID

Jie Zheng: 0000-0003-1547-3612

Author Contributions

[§]F.Y. and Y.L. contributed to this work equally.

Notes

The authors declare no competing financial interest.

ACKNOWLEDGMENTS

This work is financially supported by NSF (CBET-1510099) and National Natural Science Foundation of China (NSFC-21528601), and partially by NSF (DMR-1607475).

REFERENCES

- (1) Blaiszik, B. J.; Kramer, S. L. B.; Olugebefola, S. C.; Moore, J. S.; Sottos, N. R.; White, S. R. Self-healing polymers and composites. *Annu. Rev. Mater. Res.* **2010**, *40*, 179–211.
- (2) Gunari, N.; Brewer, L. H.; Bennett, S. M.; Sokolova, A.; Kraut, N. D.; Finlay, J. A.; Meyer, A. E.; Walker, G. C.; Wendt, D. E.; Callow, M. E.; Callow, J. A.; Bright, F. V.; Detty, M. R. The control of marine biofouling on xerogel surfaces with nanometer-scale topography. *Biofouling* **2011**, *27*, 137–149.
- (3) Ye, S.; Majumdar, P.; Chisholm, B.; Stafslie, S.; Chen, Z. Antifouling and Antimicrobial Mechanism of Tethered Quaternary Ammonium Salts in a Cross-linked Poly(dimethylsiloxane) Matrix Studied Using Sum Frequency Generation Vibrational Spectroscopy. *Langmuir* **2010**, *26*, 16455–16462.
- (4) Chen, L.; Thérien-Aubin, H.; Wong, M. C. Y.; Hoek, E. M. V.; Ober, C. K. Improved antifouling properties of polymer membranes using a “layer-by-layer” mediated method. *J. Mater. Chem. B* **2013**, *1*, 5651–5658.
- (5) Castner, D. G.; Ratner, B. D. Biomedical surface science: Foundations to frontiers. *Surf. Sci.* **2002**, *500*, 28–60.
- (6) Ekblad, T.; Bergström, G.; Ederth, T.; Conlan, S. L.; Mutton, R.; Clare, A. S.; Wang, S.; Liu, Y.; Zhao, Q.; D’Souza, F. Poly(ethylene glycol)-containing hydrogel surfaces for antifouling applications in marine and freshwater environments. *Biomacromolecules* **2008**, *9*, 2775–2783.
- (7) Chambers, L. D.; Stokes, K. R.; Walsh, F. C.; Wood, R. J. K. Modern approaches to marine antifouling coatings. *Surf. Coat. Technol.* **2006**, *201*, 3642–3652.
- (8) Banerjee, I.; Pangule, R. C.; Kane, R. S. Antifouling coatings: recent developments in the design of surfaces that prevent fouling by proteins, bacteria, and marine organisms. *Adv. Mater.* **2011**, *23*, 690–718.
- (9) Yu, Q.; Zhang, Y.; Wang, H.; Brash, J.; Chen, H. Anti-fouling bioactive surfaces. *Acta Biomater.* **2011**, *7*, 1550–1557.
- (10) Pavithra, D.; Doble, M. Biofilm formation, bacterial adhesion and host response on polymeric implants—issues and prevention. *Biomed. Mater.* **2008**, *3*, 034003.
- (11) Hetrick, E. M.; Schoenfisch, M. H. Reducing implant-related infections: active release strategies. *Chem. Soc. Rev.* **2006**, *35*, 780–789.

- (12) Knop, K.; Hoogenboom, R.; Fischer, D.; Schubert, U. S. Poly(ethylene glycol) in drug delivery: pros and cons as well as potential alternatives. *Angew. Chem., Int. Ed.* **2010**, *49*, 6288–6308.
- (13) Goda, T.; Tabata, M.; Sanjoh, M.; Uchimura, M.; Iwasaki, Y.; Miyahara, Y. Thiolated 2-methacryloyloxyethyl phosphorylcholine for an antifouling biosensor platform. *Chem. Commun.* **2013**, *49*, 8683–8685.
- (14) Sharma, S.; Johnson, R. W.; Desai, T. A. XPS and AFM analysis of antifouling PEG interfaces for microfabricated silicon biosensors. *Biosens. Bioelectron.* **2004**, *20*, 227–239.
- (15) Turner, A. Marine pollution from antifouling paint particles. *Mar. Pollut. Bull.* **2010**, *60*, 159–171.
- (16) Zhao, C.; Li, L.; Zheng, J. Achieving highly effective nonfouling performance for surface-grafted poly(HPMA) via atom-transfer radical polymerization. *Langmuir* **2010**, *26*, 17375–17382.
- (17) Ma, Y.; Dai, J.; Wu, L.; Fang, G.; Guo, Z. Enhanced anti-ultraviolet, anti-fouling and anti-bacterial polyelectrolyte membrane of polystyrene grafted with trimethyl quaternary ammonium salt modified lignin. *Polymer* **2017**, *114*, 113–121.
- (18) Ma, Y.; Lv, L.; Guo, Y.; Fu, Y.; Shao, Q.; Wu, T.; Guo, S.; Sun, K.; Guo, X.; Wujcik, E. K.; Guo, Z. Porous lignin based poly (acrylic acid)/organo-montmorillonite nanocomposites: Swelling behaviors and rapid removal of Pb (II) ions. *Polymer* **2017**, *128*, 12–23.
- (19) Ma, Y.; Sun, Y.; Fu, Y.; Fang, G.; Yan, X.; Guo, Z. Swelling behaviors of porous lignin based poly (acrylic acid). *Chemosphere* **2016**, *163*, 610–619.
- (20) Benítez, F. J.; Acero, J. L.; Leal, A. I. Application of microfiltration and ultrafiltration processes to cork processing wastewaters and assessment of the membrane fouling. *Sep. Purif. Technol.* **2006**, *50*, 354–364.
- (21) Zhao, C.; Zhao, J.; Li, X.; Wu, J.; Chen, S.; Chen, Q.; Wang, Q.; Gong, X.; Li, L.; Zheng, J. Probing structure–antifouling activity relationships of polyacrylamides and polyacrylates. *Biomaterials* **2013**, *34*, 4714–4724.
- (22) Zhang, M.; Horbett, T. A. Tetraglyme coatings reduce fibrinogen and von Willebrand factor adsorption and platelet adhesion under both static and flow conditions. *J. Biomed. Mater. Res., Part A* **2009**, *89*, 791–803.
- (23) Wahl, M. Marine epibiosis. I. Fouling and antifouling: some basic aspects. *Mar. Ecol.: Prog. Ser.* **1989**, *58*, 175–189.
- (24) Cao, B.; Tang, Q.; Li, L.; Humble, J.; Wu, H.; Liu, L.; Cheng, G. Switchable antimicrobial and antifouling hydrogels with enhanced mechanical properties. *Adv. Healthcare Mater.* **2013**, *2*, 1096–1102.
- (25) Cao, B.; Li, L.; Tang, Q.; Cheng, G. The impact of structure on elasticity, switchability, stability and functionality of an all-in-one carboxybetaine elastomer. *Biomaterials* **2013**, *34*, 7592–7600.
- (26) Cao, B.; Lee, C.-J.; Zeng, Z.; Cheng, F.; Xu, F.; Cong, H.; Cheng, G. Electroactive poly(sulfobetaine-3,4-ethylenedioxythiophene) (PSBEDOT) with controllable antifouling and antimicrobial properties. *Chem. Sci.* **2016**, *7*, 1976–1981.
- (27) Chen, S.; Li, L.; Zhao, C.; Zheng, J. Surface hydration: Principles and applications toward low-fouling/nonfouling biomaterials. *Polymer* **2010**, *51*, 5283–5293.
- (28) Chen, H.; Yang, J.; Xiao, S.; Hu, R.; Bhaway, S. M.; Vogt, B. D.; Zhang, M.; Chen, Q.; Ma, J.; Chang, Y.; Li, L.; Zheng, J. Salt-responsive polyzwitterionic materials for surface regeneration between switchable fouling and antifouling properties. *Acta Biomater.* **2016**, *40*, 62–69.
- (29) Sugawara, S.; Kodama, M.; Hirata, Y.; Kojima, S.; Yamamoto, Y. Synthesis and characterization of the most distorted 16 π porphyrin: 16 π octaisopropyltetraphenylporphyrin (OiPTPP). *J. Porphyrins Phthalocyanines* **2011**, *15*, 1326–1334.
- (30) Weinhold, F. Rebuttal to the Bickelhaupt–Baerends Case for Steric Repulsion Causing the Staggered Conformation of Ethane. *Angew. Chem., Int. Ed.* **2003**, *42*, 4188–4194.
- (31) Chen, Y.-W.; Chang, Y.; Lee, R.-H.; Li, W.-T.; Chinnathambi, A.; Alharbi, S. A.; Hsiue, G.-H. Adjustable Bioadhesive Control of PEGylated Hyperbranch Brushes on Polystyrene Microplate Interface for the Improved Sensitivity of Human Blood Typing. *Langmuir* **2014**, *30*, 9139–9146.
- (32) Wu, J.; Zhao, C.; Lin, W.; Hu, R.; Wang, Q.; Chen, H.; Li, L.; Chen, S.; Zheng, J. Binding characteristics between polyethylene glycol (PEG) and proteins in aqueous solution. *J. Mater. Chem. B* **2014**, *2*, 2983.
- (33) Volanakis, J. E.; Kaplan, M. H. Specificity of C-reactive protein for choline phosphate residues of pneumococcal C-polysaccharide. *Exp. Biol. Med.* **1971**, *136*, 612–614.
- (34) Ooi, V. C.; Liu, F. Immunomodulation and anti-cancer activity of polysaccharide-protein complexes. *Curr. Med. Chem.* **2000**, *7*, 715–729.
- (35) Yonehara, Y.; Yamashita, H.; Kawamura, C.; Itoh, K. A new antifouling paint based on a zinc acrylate copolymer. *Prog. Org. Coat.* **2001**, *42*, 150–158.
- (36) Bano, S.; Mahmood, A.; Kim, S.-J.; Lee, K.-H. Graphene oxide modified polyamide nanofiltration membrane with improved flux and antifouling properties. *J. Mater. Chem. A* **2015**, *3*, 2065–2071.
- (37) Zhao, C.; Zheng, J. Synthesis and characterization of poly(N-hydroxyethylacrylamide) for long-term antifouling ability. *Biomacromolecules* **2011**, *12*, 4071–4079.
- (38) Aldred, N.; Li, G.; Gao, Y.; Clare, A. S.; Jiang, S. Modulation of barnacle (*Balanus amphitrite* Darwin) cyprid settlement behavior by sulfobetaine and carboxybetaine methacrylate polymer coatings. *Biofouling* **2010**, *26*, 673–683.
- (39) Li, W.; Liu, Q.; Liu, L. Antifouling gold surfaces grafted with aspartic acid and glutamic acid based zwitterionic polymer brushes. *Langmuir* **2014**, *30*, 12619–12626.
- (40) Chen, H.; Zhang, M.; Yang, J.; Zhao, C.; Hu, R.; Chen, Q.; Chang, Y.; Zheng, J. Synthesis and characterization of antifouling poly(N-acryloylaminoethoxyethanol) with ultralow protein adsorption and cell attachment. *Langmuir* **2014**, *30*, 10398–10409.
- (41) Zhao, C.; Li, L.; Wang, Q.; Yu, Q.; Zheng, J. Effect of film thickness on the antifouling performance of poly(hydroxy-functional methacrylates) grafted surfaces. *Langmuir* **2011**, *27*, 4906–4913.
- (42) Yang, J.; Zhang, M.; Chen, H.; Chang, Y.; Chen, Z.; Zheng, J. Probing the structural dependence of carbon space lengths of poly(N-hydroxyalkyl acrylamide)-based brushes on antifouling performance. *Biomacromolecules* **2014**, *15*, 2982–2991.
- (43) Dai, X.; Zhang, Y.; Gao, L.; Bai, T.; Wang, W.; Cui, Y.; Liu, W. A Mechanically Strong, Highly Stable, Thermoplastic, and Self-Healable Supramolecular Polymer Hydrogel. *Adv. Mater.* **2015**, *27*, 3566–3571.
- (44) Xu, Y.; Shen, J.; Luo, X.; Zhu, W.; Chen, K.; Ma, J.; Jiang, H. Conformational transition of amyloid β -peptide. *Proc. Natl. Acad. Sci. U.S.A.* **2005**, *102*, 5403–5407.
- (45) Barcellona, M. N.; Johnson, N.; Bernards, M. T. Characterizing Drug Release from Nonfouling Polyampholyte Hydrogels. *Langmuir* **2015**, *31*, 13402–13409.
- (46) Dobbins, S. C.; McGrath, D. E.; Bernards, M. T. Nonfouling hydrogels formed from charged monomer subunits. *J. Phys. Chem. B* **2012**, *116*, 14346–14352.
- (47) Jones, D. M.; Brown, A. A.; Huck, W. T. S. Surface-Initiated Polymerizations in Aqueous Media: Effect of Initiator Density. *Langmuir* **2002**, *18*, 1265.
- (48) Vanommeslaeghe, K.; Hatcher, E.; Acharya, C.; Kundu, S.; Zhong, S.; Shim, J.; Darian, E.; Guvench, O.; Lopes, P.; Vorobyov, I.; Mackerell, A. D., Jr. CHARMM general force field: A force field for drug-like molecules compatible with the CHARMM all-atom additive biological force fields. *J. Comput. Chem.* **2010**, *31*, 671–690.
- (49) Vanommeslaeghe, K.; MacKerell, A. D., Jr. Automation of the CHARMM General Force Field (CGenFF) I: bond perception and atom typing. *J. Chem. Inf. Model.* **2012**, *52*, 3144–3154.
- (50) Vanommeslaeghe, K.; Raman, E. P.; MacKerell, A. D., Jr. Automation of the CHARMM General Force Field (CGenFF) II: assignment of bonded parameters and partial atomic charges. *J. Chem. Inf. Model.* **2012**, *52*, 3155–3168.
- (51) Yu, W.; He, X.; Vanommeslaeghe, K.; MacKerell, A. D., Jr. Extension of the CHARMM General Force Field to sulfonyl-

containing compounds and its utility in biomolecular simulations. *J. Comput. Chem.* **2012**, *33*, 2451–2468.

(52) Brooks, B. R.; Bruccoleri, R. E.; Olafson, B. D.; States, D. J.; Swaminathan, S.; Karplus, M. CHARMM: a program for macromolecular energy, minimization, and dynamics calculations. *J. Comput. Chem.* **1983**, *4*, 187–217.

(53) Kalé, L.; Skeel, R.; Bhandarkar, M.; Brunner, R.; Gursoy, A.; Krawetz, N.; Phillips, J.; Shinozaki, A.; Varadarajan, K.; Schulten, K. NAMD2: greater scalability for parallel molecular dynamics. *J. Comput. Phys.* **1999**, *151*, 283–312.

(54) MacKerell, A. D.; Bashford, D.; Bellott, M.; Dunbrack, R. L.; Evanseck, J. D.; Field, M. J.; Fischer, S.; Gao, J.; Guo, H.; Ha, S.; Joseph-McCarthy, D.; Kuchnir, L.; Kucsera, K.; Lau, F. T. K.; Mattos, C.; Michnick, S.; Ngo, T.; Nguyen, D. T.; Prodhom, B.; Reiher, W. E.; Roux, B.; Schlenkrich, M.; Smith, J. C.; Stote, R.; Straub, J.; Watanabe, M.; Wiórkiewicz-Kucsera, J.; Yin, D.; Karplus, M. All-atom empirical potential for molecular modeling and dynamics studies of proteins. *J. Phys. Chem. B* **1998**, *102*, 3586–3616.

(55) Stuart, M. A. C.; de Vos, W. M.; Leermakers, F. A. M. Why surfaces modified by flexible polymers often have a finite contact angle for good solvents. *Langmuir* **2006**, *22*, 1722–1728.

(56) Li, A.; Benetti, E. M.; Tranchida, D.; Clasohm, J. N.; Schönherr, H.; Spencer, N. D. Surface-grafted, covalently cross-linked hydrogel brushes with tunable interfacial and bulk properties. *Macromolecules* **2011**, *44*, 5344–5351.

(57) Zhang, Z.; Zhang, M.; Chen, S.; Horbett, T. A.; Ratner, B. D.; Jiang, S. Blood compatibility of surfaces with superlow protein adsorption. *Biomaterials* **2008**, *29*, 4285–4291.

(58) Dancer, S. J. The effect of antibiotics on methicillin-resistant *Staphylococcus aureus*. *J. Antimicrob. Chemother.* **2008**, *61*, 246–253.

(59) Safavieh, M.; Pandya, H. J.; Venkataraman, M.; Thirumalaraju, P.; Kanakasabapathy, M. K.; Singh, A.; Prabhakar, D.; Chug, M. K.; Shafiee, H. Rapid Real-Time Antimicrobial Susceptibility Testing with Electrical Sensing on Plastic Microchips with Printed Electrodes. *ACS Appl. Mater. Interfaces* **2017**, *9*, 12832–12840.

(60) Ren, B.; Zhang, M.; Hu, R.; Chen, H.; Wang, M.; Lin, Y.; Sun, Y.; Jia, L.; Liang, G.; Zheng, J. Identification of a new function of cardiovascular disease drug 3-morpholinopyrrolidine hydrochloride as an amyloid- β aggregation inhibitor. *ACS Omega* **2017**, *2*, 243–250.

(61) Sun, Z.; Zhang, L.; Dang, F.; Liu, Y.; Fei, Z.; Shao, Q.; Lin, H.; Guo, J.; Xiang, L.; Yerra, N.; Guo, Z. Experimental and simulation-based understanding of morphology controlled barium titanate nanoparticles under co-adsorption of surfactants. *CrystEngComm* **2017**, *19*, 3288–3298.

(62) Mark, P.; Nilsson, L. Structure and Dynamics of the TIP3P, SPC, and SPC/E Water Models at 298 K. *J. Phys. Chem. A* **2001**, *105*, 9954–9960.